‘In my beginning is my end’: Facilitating Open Scholarship and Reusability across the European Research Area

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When creating digital humanities resources – archives, projects, datasets, and multimedia– researchers have typically focused on three key questions: what technologies to use, who the audience will be, and who to involve in the creation process (Purcell, 2019; Hayler, 2016; Warwick, 2012). However, over the past decade, questions around the creation of digital scholarship have been slowly shifting. While the creation of digital archives centred around a particular individual or theme are still an important part of digital scholarship in the humanities and datasets are developed to answer specific research questions (McGillivray, et al., 2020), increasingly we are asking about solutions for long-term sustainability beyond the hope that our institutions will keep the project going (Fenlon, 2020, Smithies et al., 2019).

As Mary Queen of Scots is reputed to have embroidered onto a pillow in the years prior to her death, ‘In my end is my beginning’, or as TS Eliot reframed this in his late poem, ‘Four Quartets’: ‘In my beginning is my end’, in the digital humanities community we have come to realise that decisions made at the inception of our digital projects play an outsized role in their afterlife. While, thankfully, the vast majority of us do not need to think about our imminent demise at the end of a literal chopping block, the metaphorical one looms large for the digital scholarship we create.

The funding mechanisms we live and work within (Otis, 2023), as well as our systems of academic rewards (Hazelkorn, 2015), favours discreet scholarship with a clear beginning, middle, and end, typically culminating in a tangible product that can be made available as a type of online publication (e.g. it will remain on an institutional server for *N* number of years): a product with a web presence that can be pointed to and advertised by university marketing (Edmond and Morselli 2020) . This product-focused approach neglects the broader spectrum of valuable contributions made throughout the research process (Tasovac et al. 2023). This is why increasingly there is recognition that workflows, methodologies, and intermediate outputs are equally deserving of preservation and recognition: “it is not just the product, however incomplete, that [should be] seen as worthy of preservation for current and future generations, but also equally the process (or indeed processes) for producing it” (Viola 2023: 59).

To foster this shift in perspective and ensure that it gains wide acceptance in the research community, our funders and our institutions must realise that we are no longer operating in static, linear environments but within digital ecosystems that should prioritize both use and reuse. We are not calling for the abolition of traditional digital archives, boutique digital collections, or the individually-curated dataset. Such projects remain key to many disciplinary fields in the humanities and to researchers' own specialisations. What we are arguing for instead is the need for a fundamental change in mindset in order to design and develop digital artefacts in such a way that they integrate seamlessly into broader ecosystems of FAIR (Findable, Accessible, Interoperable and Reusable) data, services, and workflows for use and reuse (Baillot et al., 2016; Tóth-Czifra & Romary, 2020; Wilkinson, et al., 2016).

As individuals, our ability to preserve our digital scholarship is inherently limited. Yet, within new research infrastructures, digital scholarship has the possibility of living on beyond the purpose for which it was originally created. Digital Sustainability is a topic that is receiving an increasing amount of attention within the DH community (e.g. Barats, 2020; Cummings, 2020; Holmes, et al, 2023; Godard, 2019; Miller, 2022). Institutional, national, or community-based repositories serve as key mechanisms for the archiving and long-term preservation of digital artifacts (Anglada, et al., 2023; Pennock, 2024; Shearer, et al., 2023). Here data can be extracted from websites into datasets, enriched with metadata, and

stored. Even entire data-driven websites can be preserved (Milligan, Ruest, & Lin, 2016; Schafer & Winters, 2021; Vlassenroot, et al., 2019) in formats like the ISO WARC standard (BnF). Yet, even if a project team plans for sustainability and data is deposited in a repository, there is no guarantee that it finds an audience, or that even if found, the audience knows how to use it.

The paper will explore the roles that Research Infrastructures (RIs) should play in facilitating the "digital afterlife" of DH projects. We define the "digital afterlife" as the period during which DH projects as complex research outputs continue to generate scholarly, societal or cultural value after direct funding or active development come to an end. In this context, one of the most pressing questions is: what should be archived, and for how long? Should everything be preserved for the long-term, as is the case with scholarly articles in journals anticipating that scholarship will, ultimately, find its audience? Or should we take an archival approach in which the data is curated and retained (The National Archives, [n.d.]; Sabharwal, 2015; Spek & Links, 2013) based on expert judgement. These questions underscore the importance of taking a systematic approach, as has been happening across the European Research Area with the creation of research infrastructures (OECD 2023; Ribeiro 2021; Bolliger and Griffiths 2020; Edmond et al. 2020).

Research infrastructures in the Social Sciences and Humanities, such as DARIAH, CLARIN and CESSDA (ESFRI 2024), represent a significant commitment from European Member States and the EU, with substantial funds allocated to their development, maintenance, and long-term sustainability. These infrastructures offer transnational mechanisms for addressing the challenges of digital sustainability. With the increasing emphasis on Open Research, driven through initiatives such as the European Open Science Cloud (EOSC), the emphasis on scholarly reuse is only going to grow in the future. In our paper, we identify and analyze three key areas in which digital research infrastructures should play an important role in enabling and supporting the "digital afterlife" of DH projects: conceptual frameworks, technological solutions and strategic policy work.

1. **Conceptual frameworks: shifting the narrative from product to process.** Historically, DH projects have often been understood primarily as products—self-contained digital archives, datasets, or multimedia outputs. However, this product-focused perspective overlooks the iterative, collaborative, and experimental nature of much of DH work. To address this, we argue for a conceptual shift that positions process as a central element of both scholarly evaluation *and* preservation. This reframing recognizes that the workflows, methodologies, and intermediate outputs of DH projects are in some instances as valuable as the final product, often providing insights into the evolution of the research. Platforms such as the SSH Open Marketplace (Barbot et al. 2024) provide a useful framework for sharing and citing (DARIAH, 2019) these processes as first-class research outputs.
2. **Technological solutions: designing for ecosystems of use and reuse.** To support the "digital afterlife" of DH projects, it is crucial to adopt technological practices that facilitate interoperability, adaptability, and long-term sustainability from the inception of each and every project. This approach involves adopting open standards (JSON, XML, SQL, Unicode etc.); implementing modular design principles for datasets and tools; ensuring resources are accessible via APIs; providing comprehensive technical documentation; and avoiding the creation of bespoke platforms that rely on ad hoc integrations of various software packages, which makes them difficult to maintain or adapt once the original development team moves on
3. **Strategic policy work: championing sustainable DH.** To fully realize the potential of DH projects in the "digital afterlife," digital research infrastructures, within the broader context of Open Science, should actively work to reshape the funding, evaluation, and governance models that underpin DH work, emphasizing sustainability, interoperability, and long-term impact. Key strategies include advocating for funding models that prioritize long-term sustainability, encouraging the co-creation of best practices for designing interoperable digital projects, leveraging existing RIs in the SSH domain such as DARIAH, CLARIN and the broader initiatives in cultural heritage such as European Collaborative Cloud for Cultural Heritage (ECCCH) and the Common European Data Space for Cultural Heritage to establish networks of digital scholarship that transcend institutional and disciplinary silos), while developing clear guidelines for determining which data should be preserved, for how long, and under what conditions. This involves both proactive curation strategies and mechanisms for retiring datasets that no longer serve a meaningful purpose.

We argue that the successful digital afterlife of DH projects will to a large extent depend on a conceptual shift in how we conceive, design, imple-

ment and evaluate digital research. By embedding sustainability, interoperability and reuse into the very fabric of DH practices – and the design of DH projects from their inception – we should be in a much better position to ensure that the scholarly, societal and cultural value of digital projects continues long after their original funding and development have ended. Although this paper focuses on the potential of European research infrastructures to transform how we sustain and engage with DH scholarship, the model we describe may prove valuable to other countries and/or regions. However, as institutions and national initiatives decide to tackle this ever-growing issue, we are confident that the solutions can be found through the combined efforts of conceptual rethinking, technological solutions and strategic advocacy.

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